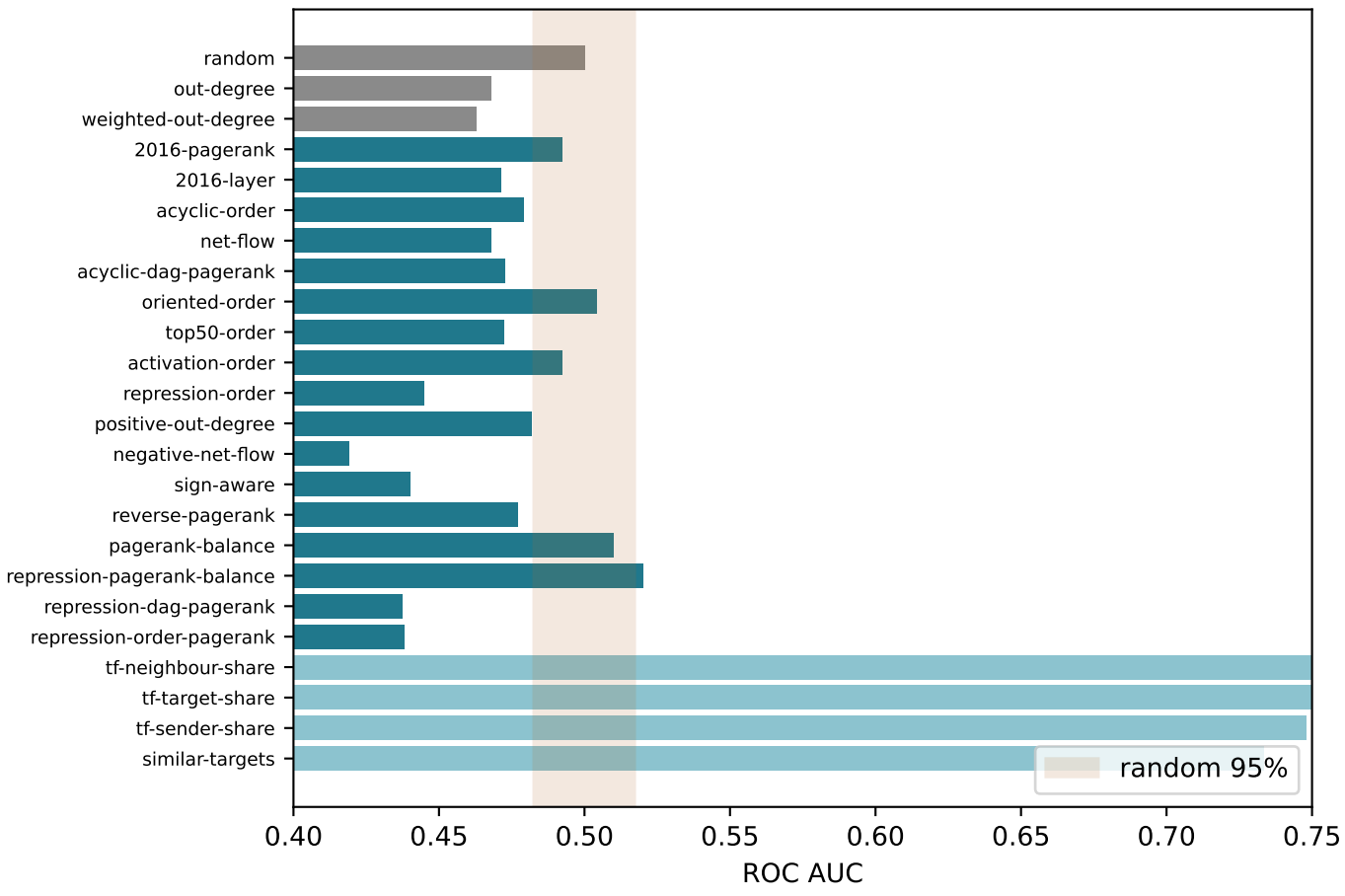
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,934 genes ranked, 1,148 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.7 (4–16)	0.500 (0.483–0.517)				
out-degree	11	0.468	0.383	1.000	+0.000 (+0.000, +0.000)	
weighted-out-degree	11	0.463	0.383	1.000	-0.005 (-0.006, -0.004)	< 0.001
2016-pagerank	8	0.492	0.768	0.815	+0.024 (+0.006, +0.042)	0.008
2016-layer	14	0.471	0.101	1.000	+0.003 (-0.013, +0.021)	0.736
acyclic-order	11	0.479	0.383	0.993	+0.011 (+0.004, +0.018)	0.002
net-flow	11	0.468	0.383	1.000	+0.000 (-0.010, +0.011)	0.974
acyclic-dag-pagerank	12	0.473	0.265	1.000	+0.005 (-0.000, +0.010)	0.072
oriented-order	12	0.504	0.265	0.330	+0.036 (+0.025, +0.048)	< 0.001
top50-order	6	0.472	0.926	1.000	+0.004 (-0.015, +0.024)	0.662
activation-order	9	0.492	0.646	0.816	+0.024 (+0.015, +0.034)	< 0.001
repression-order	10	0.445	0.509	1.000	-0.023 (-0.042, -0.003)	0.016
positive-out-degree †	7	0.482	0.862	0.981	+0.014 (+0.007, +0.020)	< 0.001
negative-net-flow †	11	0.419	0.383	1.000	-0.049 (-0.066, -0.031)	< 0.001
sign-aware †	10	0.440	0.509	1.000	-0.028 (-0.037, -0.019)	< 0.001
reverse-pagerank †	11	0.477	0.383	0.997	+0.009 (+0.004, +0.014)	0.004
pagerank-balance †	10	0.510	0.509	0.134	+0.042 (+0.027, +0.058)	< 0.001
repression-pagerank-balance †	11	0.520	0.383	0.011	+0.052 (+0.038, +0.069)	< 0.001
repression-dag-pagerank †	8	0.437	0.768	1.000	-0.031 (-0.048, -0.011)	< 0.001
repression-order-pagerank †	11	0.438	0.383	1.000	-0.030 (-0.049, -0.011)	< 0.001
tf-neighbour-share ‡	92	0.770	< 0.001	< 0.001	+0.302 (+0.278, +0.326)	< 0.001
tf-target-share ‡	83	0.767	< 0.001	< 0.001	+0.299 (+0.276, +0.322)	< 0.001
tf-sender-share ‡	90	0.748	< 0.001	< 0.001	+0.280 (+0.256, +0.304)	< 0.001
similar-targets ‡	88	0.734	< 0.001	< 0.001	+0.266 (+0.242, +0.291)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,837	0.045 (2.86×)	3.93×	1.14×	0.77 (0.77)
resolve_reciprocals → drop_sinks → prune_triangles	31,789	0.008 (1.25×)	3.16×	1.15×	0.21 (0.21)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	18,114	0.011 (0.92×)	2.44×	0.90×	0.14 (0.14)
absolute	31,837	0.045 (2.86×)			
absolute → minimum_feedback_order	31,698	0.031 (2.47×)			
orient_reciprocals ratio=2.0 → absolute	31,825	0.024 (2.12×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	31,724	0.021 (1.98×)			
top_outgoing k=50 → absolute	31,837	0.015 (4.23×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,675	0.010 (3.75×)			
keep_sign sign=1 → absolute	31,837	0.038 (3.93×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,806	0.028 (4.02×)			
keep_sign sign=-1 → absolute	25,120	0.009 (1.09×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	23,903	0.009 (1.03×)			
keep_sign sign=-1	25,120	0.009 (1.09×)		1.09×	0.13 (0.13)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,028,167 edges (665,924 negative), largest strong component 11,914.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,028,167 edges (665,924 negative), largest strong component 11,914.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 674,641 edges (632,829 negative), largest strong component 11,716.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 674,641 edges (632,829 negative), largest strong component 11,716; DAG keeps 547,728 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,028,167 edges (0 negative), largest strong component 11,914; DAG keeps 1,545,294 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,028,167 edges (0 negative), largest strong component 11,914.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,028,167 edges (0 negative), largest strong component 11,914; DAG keeps 1,545,294 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,346,942 edges (0 negative), largest strong component 11,707; DAG keeps 1,238,247 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 513,896 edges (0 negative), largest strong component 11,899; DAG keeps 355,190 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,362,243 edges (0 negative), largest strong component 11,875; DAG keeps 958,958 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 665,924 edges (0 negative), largest strong component 8,147; DAG keeps 633,027 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,028,167 edges (665,924 negative), largest strong component 11,914.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,028,167 edges (665,924 negative), largest strong component 11,914.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,028,167 edges (665,924 negative), largest strong component 11,914.

reverse-pagerank: Highest reverse PageRank (raw graph)